

An Empirical Analysis of Brute Force Authentication Attacks and Preventive Controls

Group – 9

1. Abstract

Brute force attacks remain one of the most persistent and effective methods used to compromise authentication systems. This project investigates the behavior and impact of brute force attacks on SSH-based authentication in a controlled cloud environment using Microsoft Azure. A deliberately vulnerable system is configured and subjected to password-based attacks using automated tools. The study then evaluates the effectiveness of preventive controls such as strong passwords and Fail2Ban. Results demonstrate that weak credentials are highly susceptible to compromise, while simple defensive mechanisms significantly reduce attack success rates.

2. Introduction

Authentication systems form the first line of defense in any computing environment. Despite advancements in security, password-based authentication remains widely used and frequently misconfigured. Brute force attacks exploit weak credentials and lack of protective controls by systematically attempting multiple password combinations.

This project focuses on:

- Simulating brute force attacks in a controlled Azure environment
- Observing system behavior through logs
- Evaluating preventive measures

The objective is to provide empirical evidence of both vulnerability and mitigation effectiveness

3. Methodology

3.1 Environment Setup

The experiment was conducted using Microsoft Azure with the following architecture:

- Virtual Network: 10.0.0.0/16
- Subnets:

- Attacker Subnet: 10.0.1.0/24
- Target Subnet: 10.0.2.0/24

3.2 Virtual Machines

- **Attacker VM**
 - Ubuntu Server 24.04
 - Public IP access
 - Used to launch brute force attacks
- **Target VM**
 - Ubuntu Server 24.04
 - Public IP access
 - Hosted SSH service

3.3 Security Configuration

Network Security Groups were configured to enforce strict access control:

- SSH access allowed only from:
 - Admin public IP
 - Attacker Public IP (98.70.28.75)
- All other inbound traffic denied

This ensured a controlled and realistic attack surface.

4. Implementation

4.1 Target System Configuration

The target system was intentionally weakened:

- SSH password authentication enabled
- User created:
 - Username: vulnvictim
 - Password: 12345

This setup simulates poor security practices commonly found in real systems.

4.2 Attack Setup

The attacker VM was configured with Hydra and a commonly used password list (wordlist.txt). Connectivity between attacker and target was verified within the private network.

4.3 Brute Force Attack Execution

The following command was used from the Attacker Machine:

```
hydra -l victim -P wordlist.txt 20.244.24.0 ssh -t 4
```

This command systematically attempts passwords from the wordlist against the SSH service.

4.4 Observations During Attack

Real-time monitoring of logs on the target system showed:

- Repeated failed login attempts
- High-frequency authentication requests
- Consistent source IP from attacker VM

Example log pattern:

Failed password for victim from 98.70.28.75 port 22 ssh2

5. Preventive Controls

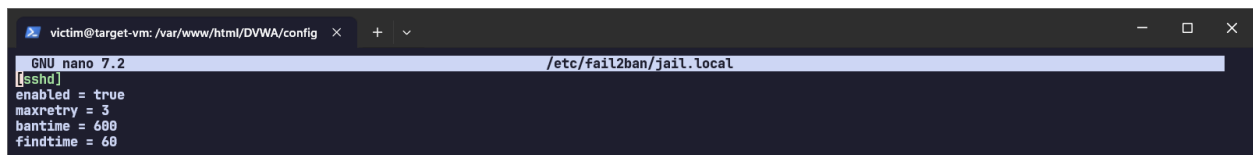
5.1 Strong Password Enforcement

The weak password was replaced with a complex password.
Result:

- Brute force attack failed to find a match
- Time and attempts increased significantly

5.2 Fail2Ban Implementation

Fail2Ban was configured with:

A screenshot of a terminal window. The title bar shows 'victim@target-vm: /var/www/html/DVWA/config' and standard window controls. The terminal content shows the 'nano' editor editing '/etc/fail2ban/jail.local'. The configuration for '[sshd]' is visible: 'enabled = true', 'maxretry = 3', 'bantime = 600', and 'findtime = 60'.

```
victim@target-vm: /var/www/html/DVWA/config  x  +  v
GNU nano 7.2 /etc/fail2ban/jail.local
[sshd]
enabled = true
maxretry = 3
bantime = 600
findtime = 60
```

- maxretry = 3
- bantime = 600
- findtime = 60

Behavior observed:

- After multiple failed attempts, attacker IP was banned
- Subsequent connection attempts were blocked

This demonstrated effective automated defense against brute force attempts.

6. Experimental Results

Scenario	Password Strength	Protection	Attempts	Time	Outcome
1	Weak (12345)	None	Low	Fast	Success
2	Strong	None	High	Long	Failure
3	Weak	Fail2Ban	Limited	Short	Blocked

7. Analysis

The experiment highlights several key findings:

- Weak passwords are the primary vulnerability exploited by brute force attacks
- Attack tools like Hydra can compromise such systems rapidly
- Log analysis provides clear indicators of attack patterns
- Preventive controls drastically reduce attack effectiveness

Fail2Ban proved particularly effective by:

- Detecting repeated failures
- Automatically blocking malicious IPs

```
victim@target-vm:/var/www/html/DVWA/config$ sudo fail2ban-client status sshd
Status for the jail: sshd
- Filter
|   - Currently failed: 0
|   - Total failed: 0
|   - Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
- Actions
|   - Currently banned: 1
|   - Total banned: 1
|   - Banned IP list: 98.70.28.75
```

Even without advanced security systems, basic controls significantly enhance system resilience.

8. Conclusion

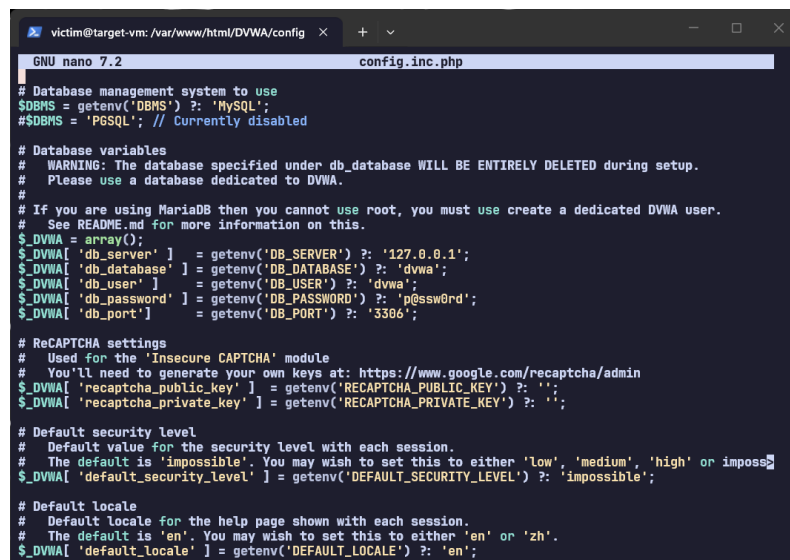
This project demonstrates that brute force attacks are both simple and highly effective against poorly configured systems. However, it also shows that equally simple defenses—such as strong passwords and automated banning mechanisms—can mitigate these threats effectively.

The study reinforces the importance of:

- Proper authentication practices
- Monitoring and logging
- Proactive defensive configurations

9. Limitations

- Only SSH-based brute force attacks were deeply analyzed
- Web-based attacks were minimally explored through dvwa configuration:



```
victim@target-vm: /var/www/html/DVWA/config x + v
GNU nano 7.2 config.inc.php

# Database management system to use
$DBMS = getenv('DBMS') ? 'MySQL':
#$DBMS = 'PGSQL'; // Currently disabled

# Database variables
# WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
# Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
# See README.md for more information on this.
$_DVWA = array();
$_DVWA['db_server'] = getenv('DB_SERVER') ? '127.0.0.1':
$_DVWA['db_database'] = getenv('DB_DATABASE') ? 'dvwa':
$_DVWA['db_user'] = getenv('DB_USER') ? 'dvwa':
$_DVWA['db_password'] = getenv('DB_PASSWORD') ? 'p@ssw0rd':
$_DVWA['db_port'] = getenv('DB_PORT') ? '3306';

# ReCAPTCHA settings
# Used for the 'Insecure CAPTCHA' module
# You'll need to generate your own keys at: https://www.google.com/recaptcha/admin
$_DVWA['recaptcha_public_key'] = getenv('RECAPTCHA_PUBLIC_KEY') ? '':
$_DVWA['recaptcha_private_key'] = getenv('RECAPTCHA_PRIVATE_KEY') ? '':

# Default security level
# Default value for the security level with each session.
# The default is 'impossible'. You may wish to set this to either 'low', 'medium', 'high' or impos
$_DVWA['default_security_level'] = getenv('DEFAULT_SECURITY_LEVEL') ? 'impossible';

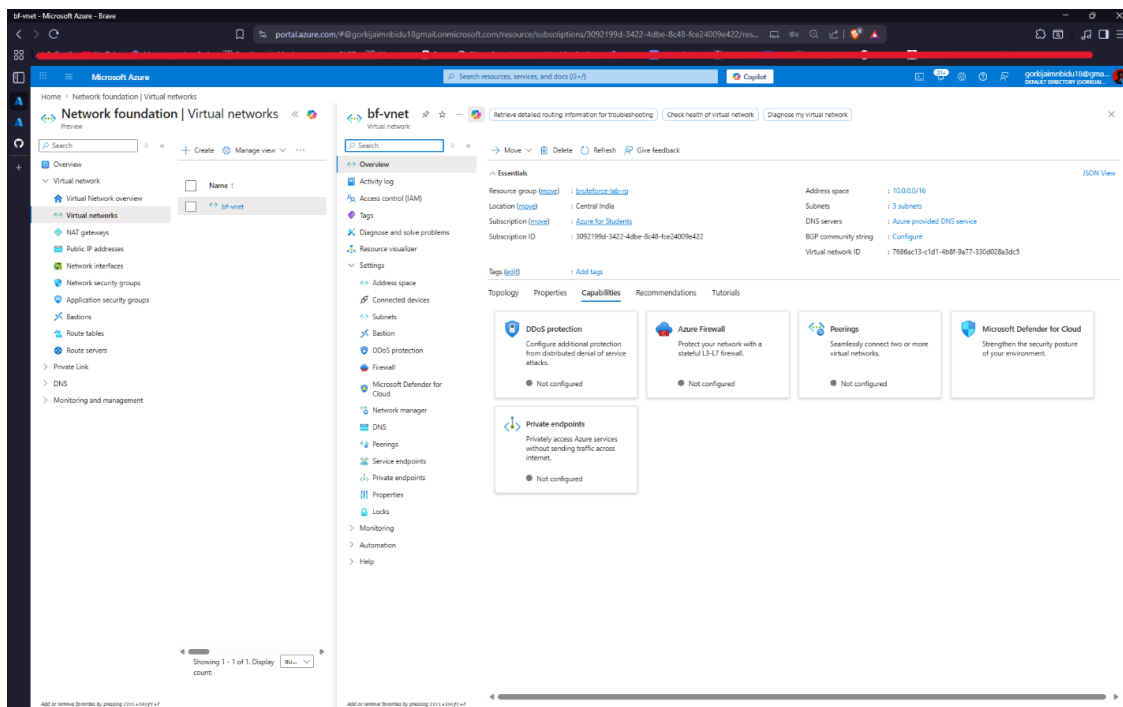
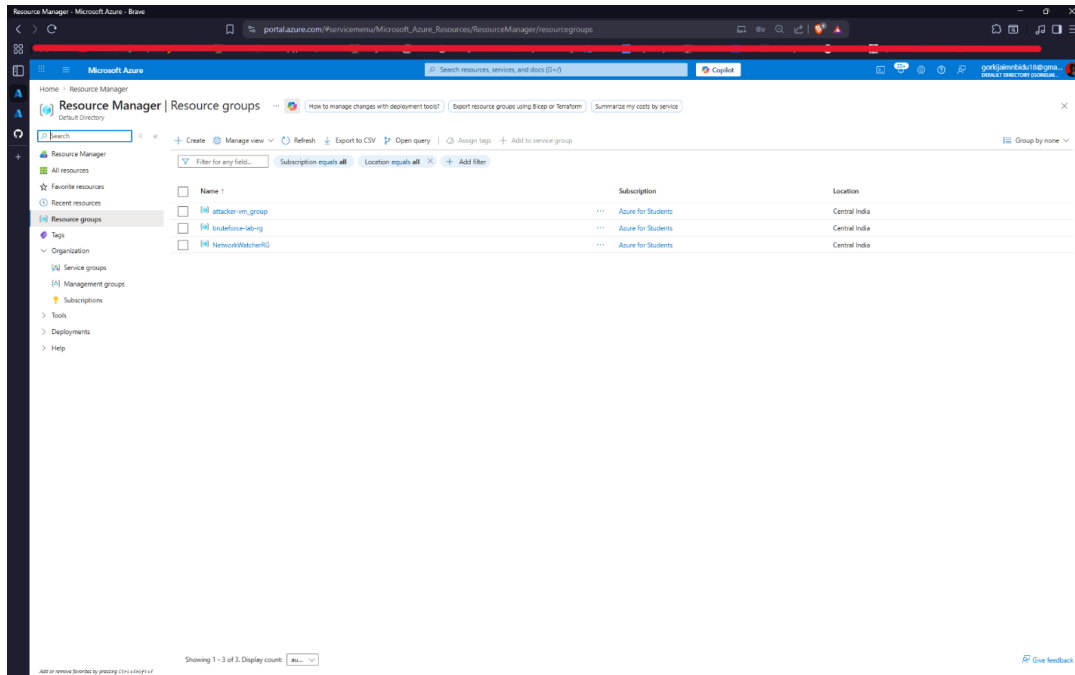
# Default locale
# Default locale for the help page shown with each session.
# The default is 'en'. You may wish to set this to either 'en' or 'zh'.
$_DVWA['default_locale'] = getenv('DEFAULT_LOCALE') ? 'en';
```

- Advanced detection systems were not included(WAZUH SIEM or ELK stack)

10. Screenshots of the practice

Azure Setup

- Virtual Network and subnet configuration



- VM creation summary

The screenshot displays the Microsoft Azure portal interface for the 'attacker-vm' virtual machine. The left sidebar shows the navigation menu with 'Virtual machines' selected. The main content area is divided into two panels: 'Overview' on the left and 'Essentials' on the right.

Overview Panel:

- Activity log:** Shows the VM's status as 'Running'.
- Access control (IAM):** Shows the VM's location as 'Central India (Zone 1)' and its subscription as '30921996-3422-4d8e-8c48-fce24009e422'.
- Tags:** Shows the VM's tags, including 'attacker-vm'.
- Network settings:** Shows the VM's network configuration, including the 'attacker-vm' network interface.
- Disks:** Shows the VM's disk configuration, including the 'attacker-vm' disk.

Essentials Panel:

- Resource group:** bruceforce-lab-rg
- Status:** Running
- Location:** Central India (Zone 1)
- Subscription:** 30921996-3422-4d8e-8c48-fce24009e422
- Availability zone:** 1
- Operating system:** Linux (ubuntu 24.04)
- Size:** Standard B2as v2 (2 vcpus, 8 GiB memory)
- Primary NIC public IP:** 98.70.28.75
- Virtual network/subnet:** bf-vnet/attacker-sub
- DNS name:** Not configured
- Health state:** -
- Time created:** 4/17/2026, 11:48 AM UTC

Properties Panel:

- Virtual machine:**
 - Computer name: attacker-vm
 - Operating system: Linux (ubuntu 24.04)
 - VM generation: V2
 - VM architecture: x64
 - Agent status: Ready
 - Agent version: 2.15.1.3
 - Hibernation: Disabled
 - Host group: -
 - Host: -
 - Proximity placement group: -
 - Colocation status: N/A
 - Capacity reservation group: -
 - Disk controller type: SCSI
- Azure Spot:**
 - Azure Spot: -
 - Azure Spot eviction policy: -
- Availability + scaling:**
 - Availability zone: 1
 - Extended zone: -
 - Availability set: -

Networking Panel:

- Public IP address:** 98.70.28.75 (Network interface attacker-vm644_x1)
- Public IP address (IPv6):** -
- Private IP address:** 100.1.5
- Private IP address (IPv6):** -
- Virtual network/subnet:** bf-vnet/attacker-sub
- DNS name:** Configure

Size Panel:

- Size:** Standard B2as v2
- vCPUs:** 2
- RAM:** 8 GiB
- Threads per core:** 2

Source image details Panel:

- Source image publisher:** canonical
- Source image offer:** ubuntu-24.04-lts
- Source image plan:** server

Disk Panel:

- OS disk:** attacker-vm_disk1_f29b8d3e1446b0b6c3d65e2a0dcfc
- Encryption at host:** Disabled
- Azure disk encryption:** Not enabled

The screenshot displays the Microsoft Azure portal interface for the 'target-vm' virtual machine. The left sidebar shows the navigation menu with 'Virtual machines' selected. The main content area is divided into two panels: 'Overview' on the left and 'Essentials' on the right.

Overview Panel:

- Activity log:** Shows the VM's status as 'Running'.
- Access control (IAM):** Shows the VM's location as 'Central India (Zone 1)' and its subscription as '30921996-3422-4d8e-8c48-fce24009e422'.
- Tags:** Shows the VM's tags, including 'target-vm'.
- Network settings:** Shows the VM's network configuration, including the 'target-vm' network interface.
- Disks:** Shows the VM's disk configuration, including the 'target-vm' disk.

Essentials Panel:

- Resource group:** bruceforce-lab-rg
- Status:** Running
- Location:** Central India (Zone 1)
- Subscription:** 30921996-3422-4d8e-8c48-fce24009e422
- Availability zone:** 1
- Operating system:** Linux (ubuntu 24.04)
- Size:** Standard B2as v2 (2 vcpus, 8 GiB memory)
- Primary NIC public IP:** 20.244.24.0
- Virtual network/subnet:** bf-vnet/target-sub
- DNS name:** Not configured
- Health state:** -
- Time created:** 4/17/2026, 11:22 AM UTC

Properties Panel:

- Virtual machine:**
 - Computer name: target-vm
 - Operating system: Linux (ubuntu 24.04)
 - VM generation: V2
 - VM architecture: x64
 - Agent status: Ready
 - Agent version: 2.15.1.3
 - Hibernation: Disabled
 - Host group: -
 - Host: -
 - Proximity placement group: -
 - Colocation status: N/A
 - Capacity reservation group: -
 - Disk controller type: SCSI
- Azure Spot:**
 - Azure Spot: -
 - Azure Spot eviction policy: -
- Availability + scaling:**
 - Availability zone: 1
 - Extended zone: -
 - Availability set: -

Networking Panel:

- Public IP address:** -
- Public IP address (IPv6):** -
- Private IP address:** 100.2.5
- Private IP address (IPv6):** -
- Virtual network/subnet:** bf-vnet/target-sub
- DNS name:** Configure

Size Panel:

- Size:** Standard B2as v2
- vCPUs:** 2
- RAM:** 8 GiB
- Threads per core:** 2

Source image details Panel:

- Source image publisher:** canonical
- Source image offer:** ubuntu-24.04-lts
- Source image plan:** server

Disk Panel:

- OS disk:** target-vm_disk1_635a5ed599494d74c17560373f6e3
- Encryption at host:** Disabled
- Azure disk encryption:** Not enabled
- Ephemeral OS disk:** N/A

Access & Connectivity

- SSH login to target VM

```
victim@target-vm: /var/www/html/DVWA/config x + v
PS C:\Users\gorki\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Windows PowerShell> ssh victim@20.244.24.0
The authenticity of host '20.244.24.0 (20.244.24.0)' can't be established.
ED25519 key fingerprint is SHA256:2ayghKxhZY30IKg0L3Wehqvd0LU09YqXhY6z5I9uU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.244.24.0' (ED25519) to the list of known hosts.
victim@20.244.24.0's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1011-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Apr 17 11:32:15 UTC 2026

System load: 0.0          Processes: 132
Usage of /:  5.7% of 28.02GB    Users logged in: 0
Memory usage: 3%             IPv6 address for eth0: 10.0.2.5
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software:
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

victim@target-vm:~$
victim@target-vm:~$
```

Attack Execution

- Hydra command running and gaining access

```
attacker@attacker-vm: ~ x + v
attacker@attacker-vm:~$
attacker@attacker-vm:~$ hydra -l vulnvictim -P wordlist.txt 20.244.24.0 ssh -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is no
n-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-17 12:01:57
[DATA] max 4 tasks per 1 server, overall 4 tasks, 209 login tries (l:1/p:209), ~53 tries per task
[DATA] attacking ssh://20.244.24.0:22/
[22][ssh] host: 20.244.24.0 login: vulnvictim password: 12345
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-17 12:02:02
attacker@attacker-vm:~$
```

Logs

- auth.log showing repeated failed attempts suggesting clear brute force attempt

```
victim@target-vm: /var/www/html/DVWA/config
victim@target-vm: /var/www/html/DVWA/config$
victim@target-vm: /var/www/html/DVWA/config$ ip ad
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0d:3a:f0:b3:a2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.5/24 metric 100 brd 10.0.2.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20d:3aff:fe0:b3a2/64 scope link
        valid_lft forever preferred_lft forever
3: enP48624s1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master eth0 state UP group default qlen 1000
    link/ether 00:0d:3a:f0:b3:a2 brd ff:ff:ff:ff:ff:ff
    altname enP48624p0s2
victim@target-vm: /var/www/html/DVWA/config$ sudo tail -f /var/log/auth.log
2026-04-17T12:02:02.245951+00:00 target-vm sshd[18975]: Failed password for vulnictim from 98.70.28.75 port 42622 ssh2
2026-04-17T12:02:02.252851+00:00 target-vm sshd[18974]: Failed password for vulnictim from 98.70.28.75 port 42688 ssh2
2026-04-17T12:02:02.259954+00:00 target-vm sshd[18973]: Failed password for vulnictim from 98.70.28.75 port 42596 ssh2
2026-04-17T12:02:02.503915+00:00 target-vm sshd[18974]: Connection closed by authenticating user vulnictim 98.70.28.75 port 42688 [preauth]
2026-04-17T12:02:02.504259+00:00 target-vm sshd[18974]: PAM 1 more authentication failure; Logname= uid=0 euid=0 tty=ssh ruser= rhost=98.70.28.75 user=vulnictim
2026-04-17T12:02:02.513838+00:00 target-vm sshd[18973]: Connection closed by authenticating user vulnictim 98.70.28.75 port 42596 [preauth]
2026-04-17T12:02:02.514283+00:00 target-vm sshd[18975]: Connection closed by authenticating user vulnictim 98.70.28.75 port 42622 [preauth]
2026-04-17T12:02:02.514396+00:00 target-vm sshd[18973]: PAM 1 more authentication failure; Logname= uid=0 euid=0 tty=ssh ruser= rhost=98.70.28.75 user=vulnictim
2026-04-17T12:02:02.514445+00:00 target-vm sshd[18975]: PAM 1 more authentication failure; Logname= uid=0 euid=0 tty=ssh ruser= rhost=98.70.28.75 user=vulnictim
2026-04-17T12:02:48.847346+00:00 target-vm sudo: victim : TTY=pts/0 ; PWD=/var/www/html/DVWA/config ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-04-17T12:02:48.852424+00:00 target-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by victim(uid=1000)
```

Fail2Ban

- Fail2Ban status output

```
attacker@attacker-vm: ~
Syntax: hydra [[[-l LOGIN|-L FILE] [-p PASS|-P FILE]] | [-C FILE]] [-e nsr] [-o FILE] [-t TASKS] [-M FILE [-T TASKS]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX:CHARSET] [-c TIME] [-i ISOuvVd46] [-m MODULE_OPT] [service://server[:PORT]/OPT]]

Options:
-l LOGIN or -L FILE login with LOGIN name, or load several logins from FILE
-p PASS or -P FILE try password PASS, or load several passwords from FILE
-C FILE colon separated "login:pass" format, instead of -L/-P options
-M FILE list of servers to attack, one entry per line, ':' to specify port
-t TASKS run TASKS number of connects in parallel per target (default: 16)
-U service module usage details
-m OPT options specific for a module, see -U output for information
-h more command line options (COMPLETE HELP)
server the target: DNS, IP or 192.168.0.0/24 (this OR the -M option)
service the service to crack (see below for supported protocols)
OPT some service modules support additional input (-U for module help)

Supported services: adam6500 asterisk cisco cisco-enable cobaltstrike cvs firebird ftp[s] http[s]-{head|get|post} http[s]-{get|post}-form http-proxy http-p
roxy-urlenum icq imap[s] irc ldap2[s] ldap3[-{cram|digest}|md5][s] memcached mongodb mssql mysql nntp oracle-listener oracle-sid pcanwhere pcnfs pop3[s] po
stgres radmin2 rdp redis rexec rlogin rpcap rsh rtsp s7-300 sip smb smtp[s] smtp-enum snmp socks5 ssh sshkey svn teamspeak telnet[s] vmauthd vnc xmpp

Hydra is a tool to guess/crack valid login/password pairs.
Licensed under AGPL v3.0. The newest version is always available at:
https://github.com/vanhauser-thc/thc-hydra
Please don't use in military or secret service organizations, or for illegal
purposes. (This is a wish and non-binding - most such people do not care about
laws and ethics anyway - and tell themselves they are one of the good ones.)

Example: hydra -l user -P passlist.txt ftp://192.168.0.1

real 0m0.014s
user 0m0.008s
sys 0m0.006s
```

- Banned IP entry

```
victim@target-vm: /var/www/html/DVWA/config
""{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-nft-ap\s+.*\s*\Khandle\s+(\d+)$'; } | while read -r hdl; do"",
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
Setting up python3-pyinotify (0.9.6-2ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victim@target-vm: /var/www/html/DVWA/config$ sudo nano /etc/fail2ban/jail.local
victim@target-vm: /var/www/html/DVWA/config$
victim@target-vm: /var/www/html/DVWA/config$
victim@target-vm: /var/www/html/DVWA/config$ sudo systemctl restart fail2ban
victim@target-vm: /var/www/html/DVWA/config$ sudo fail2ban-client status sshd
Status for the jail: sshd
- Filter
  - Currently failed: 0
  - Total failed: 0
  - Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
- Actions
  - Currently banned: 1
  - Total banned: 1
  - Banned IP list: 98.70.28.75
victim@target-vm: /var/www/html/DVWA/config$ sudo fail2ban-client status sshd
Status for the jail: sshd
- Filter
  - Currently failed: 0
  - Total failed: 0
  - Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
- Actions
  - Currently banned: 1
  - Total banned: 1
  - Banned IP list: 98.70.28.75
victim@target-vm: /var/www/html/DVWA/config$
```

Post-Defense

- Hydra failing after protection

```
attacker@attacker-vm: ~
Syntax: hydra [[-l LOGIN|-L FILE] [-p PASS|-P FILE]] [-c FILE] [-e nsr] [-o FILE] [-t TASKS] [-M FILE [-T TASKS]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX:CHARSET] [-c TIME] [-I ISOuvVd46] [-m MODULE_OPT] [service://server[:PORT]]/[OPT]]

Options:
-l LOGIN or -L FILE login with LOGIN name, or load several logins from FILE
-p PASS or -P FILE try password PASS, or load several passwords from FILE
-c FILE colon separated "login:pass" format, instead of -L/-P options
-M FILE list of servers to attack, one entry per line, "." to specify port
-t TASKS run TASKS number of connects in parallel per target (default: 16)
-u service module usage details
-m OPT options specific for a module, see -U output for information
-h more command line options (COMPLETE HELP)
server the target: DNS, IP or 192.168.0.0/24 (this OR the -M option)
service the service to crack (see below for supported protocols)
OPT some service modules support additional input (-U for module help)

Supported services: adam6500 asterisk cisco cisco-enable cobaltstrike cvs firebird ftp[s] http[s]-{head|get|post} http[s]-{get|post}-form http-proxy http-p
roxy-urlenum icq imap[s] irc ldap2[s] ldap3[-{cram|digest}|md5][s] memcached mongodb mssql mysql nntp oracle-listener oracle-sid pcanywhere pcnfs pop3[s] po
stgres radmin2 rdp redis rexec rlogin rpcap rsh rtp s7-300 sip smb smtp[s] smtp-enum snmp socks5 ssh sshkey svn teamspeak telnet[s] vmauthd vnc xmpp

Hydra is a tool to guess/crack valid login/password pairs.
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Example: hydra -l user -P passlist.txt ftp://192.168.0.1

real 0m0.014s
user 0m0.008s
sys 0m0.004s
attacker@attacker-vm:~$ hydra -l vulnictim -P wordlist.txt 20.244.24.0 ssh -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is no
n-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-17 12:04:23
[DATA] max 4 tasks per 1 server, overall 4 tasks, 289 login tries (1:1/p:289), ~53 tries per task
[DATA] attacking ssh://20.244.24.0:22/
[ERROR] could not connect to ssh://20.244.24.0:22 - Timeout connecting to 20.244.24.0
attacker@attacker-vm:~$
```